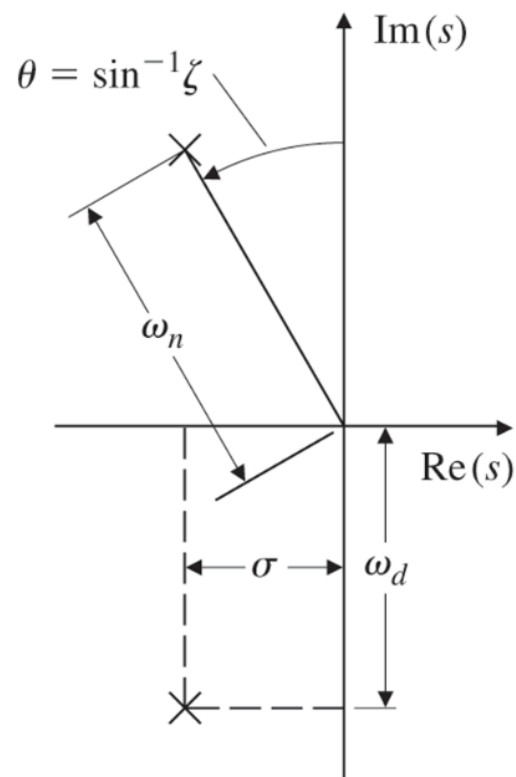
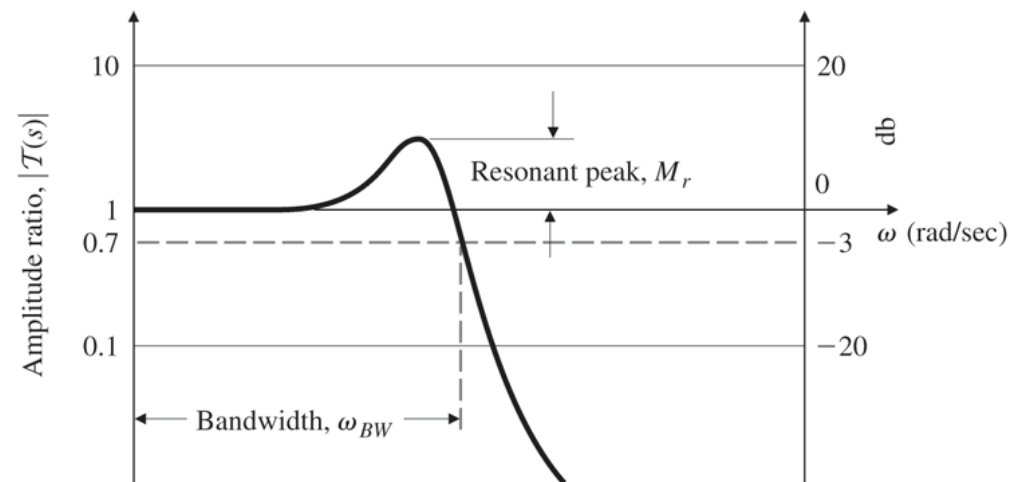
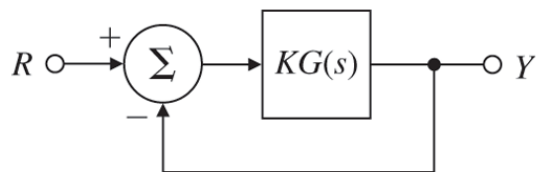
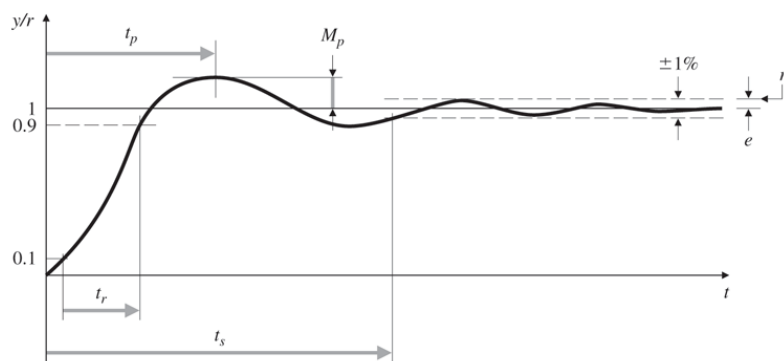


Design Aids

Closed Loop



$$t_s = \frac{4.6}{\sigma}$$

$$t_r = \frac{1.8}{\omega_n}$$

$$\sigma = \zeta \omega_n$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$e_{ss} = \frac{1}{1 + K_0}, \quad K_0 = |G(j\omega)|_{\omega=0}$$

$$|E| < \frac{1}{1 + W_1}, \quad \omega < \omega_1$$

$$\omega_{BW} = \omega_c \quad \text{for } PM = 90^\circ$$

$$\omega_{BW} = 2\omega_c \quad \text{for } PM = 45^\circ$$

$$M_r \cong \frac{1}{2 \sin(PM/2)}$$

$$M_p = 5\%, \quad \zeta = 0.7$$

$$M_p = 15\%, \quad \zeta = 0.5$$

$$M_p = 35\%, \quad \zeta = 0.3$$

$$\zeta \cong \frac{PM}{100} \quad \text{for } PM < 70^\circ$$